

1) Version Control System:-

Git:-

→ Git is a version control system where the user can write, run & debug the program. It allows the users to collaborate, maintain the source code.

* Git allows the user to track the changes that happened over a period time.

→ It allows the user to collaborate, maintain the code data.

Git Hub:-

→ Git Hub is a cloud-based platform where it allows the user to use it as a Host for the Git.

* It allows the users to maintain data privacy.

2. Programming Language & Development Environment:

Python:-

* Python is a high-level general purpose language used to develop web applications, apps and widely used in Data Science.

* Readability: - English-based syntax.

* You don't need to declare the variable. It can be determined ~~with~~ during Run Time.

* Example:- `print("Hello World")`

SQL:-

* SQL stands for Structured Query Language used to manage & maintain the RDBMS (Relational Database).

* The Table Based - means the data are arranged in Rows & column.

* You need to specify what to get, not how not to get it

* Example:- `select * from`

`STUDENT`

`where GRADE "A"`

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b) Key elements of Programming languages.

Syntax: There is a certain rule that need to be followed while writing codes.

Ex)

```
{  
  {  
    a=2, b=4.  
    c=a+b  
  }  
}
```

Follow

~~FOR LOOP~~

The opening & the closing in the Brackets should be in the same line.

Variables:-

These are the alphabet & character that stores value. It is very important in R, without the declaration ~~pro~~ R cannot fetch details.

Ex) a=20 x=80
 └─ variable

Loops:-

* Loops are the special function in programming language like python, C++ etc. It allows the user to simplify the program and avoid repeating the steps.

* IF, FOR, ~~IF~~ IF ELSE

Ex)

```
( if a>b  
  printf("A is greater")  
  Else  
  printf("B is greater")
```


c) Docker,

* Docker is a development Environment where it allows the user to write, Run and Debug the program.

* It allows the programmer to Run (or) deploy before they go live.

* The user can check, change the code or program before deployment

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d) Concept of IDE,

* IDE stands for Integrated Development Environment

* In this, all the Data, App Analysing, Deploy environ^{en} are integrated.

* ~~The~~ The person from one computer can work on the other codes if he needs to alter.

* RStudio, ^{Docker} ~~RStudio~~, Jupyter are popular IDE used for Data Analytics.

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MBA90, 02

Studiengang / Fachsemester Study program / Subject semester

Analytical Skills for Business

Prüfungsfach / Prüfungsdozent:in Examination subject / Examiner

a) 3) No-code and Low code Tools.

Snowflake :-

* No-code and Low code Tools allows to the user to ~~build~~ build a data using minimal or no-code.

* Snowflake allow the users to work on project with the minimal code and it can be linked with GitHub.

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* It help the user to exchange the code and analyse the value with minimal ~~effort~~ code.

b) Advantages of using no-code/Low code tools in Business.

* With the help of No-code/Low code any one can design, change the program ~~be~~ if he/she knows about how the data works in Business Analytics.

* The user can just type the details he wants and the AI that is integrated with the tools will provide the ^{detail} details.

→ These tools save time and energy and allows the researchers to focus on their Role.

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4. Statistics and Analytics

a) Measure of Central Tendency

→ Descriptive Statistics is numerical and graphical tool that summarize the main dataset showing its Central Tendency (mean, median, mode) and variability (Range, variance).

Central Tendency

Mean = The total number of items divided by the total counts.

$$\begin{aligned} & \text{EX: } \cancel{1+2+3+4} \quad \text{MEAN} = \frac{1+2+3+4}{4} = 10 \div 4 \end{aligned}$$

Example:-

$$A = 0, 1, 2, 3, 4$$

MEAN

$$\begin{aligned} \text{MEAN} &= 0+1+2+3+4/5 \\ &= 10/5 \\ &= 2 \end{aligned}$$

→ when the Data Analyst wants to find the mean of the Salary. It can be useful.

Median:

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It is the middle character of a dataset.

ODD:

$$A = 1, 2, 3, 4, 5$$

$$\text{MEDIAN} = 3$$

If Even,

$$A = 1, 2, \overbrace{3, 3}, 5, 6$$

$$= 3 + 3/2$$

$$= 6/2 = 3 \rightarrow \text{Median}$$

Mode:

It is the most repetitive numbers that occurs in a dataset.

$$A = 1, 2, \overbrace{3, 4, 5, 6}, 3$$

$$\text{mode} = 3$$

When there are 2 or more same characters are provided. This can be used to identify the repetitive numbers.

b) Univariate

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5.) Practical R Programming:-

b) Get current R working directory

`GetWD()`

`SetWD("C:Documents/WorkingR")`

c) R code to access help documentation.

`?mean()`

a) R code to calculate & store the variable.

$$3 + \sqrt{9} - 2^2 \cdot \frac{4}{3}$$

Result

$$= (3 + \text{SQRT}(9) - 2^2) * (4 \div 3)$$

d)

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